TESLA COACHING CLASSES

Science and Technology – Part 1

Time: 2 Hours	**Standard: Class 10**	**Total Marks: 40**

Part Syllabus Test - 06 (PST-06)

Student Name: _____ **Date:** _____

CHAPTERS COVERED & WEIGHTAGE

CHAPTER 1	Gravitation	5%
CHAPTER 2	Periodic Classification of Elements	5%
CHAPTER 3	Chemical Reactions and Equations	80%
CHAPTER 6	Refraction of Light	5%
CHAPTER 7	Lenses	5%

Q.1(A) **Choose the correct alternative and rewrite the sentences:** **[5 Marks]**

i) The chemical reaction in which two or more products are formed from a single reactant is called _____ reaction.

(a) decomposition **(b)** combination

(c) displacement **(d)** double displacement

ii) _____ is in liquid form in the halogen family.

(a) Fluorine **(b)** Chlorine

(c) Bromine **(d)** Iodine

iii) A ray of light gets refracted _____ while entering the lens.

(a) once **(b)** twice

(c) thrice **(d)** doesn't happen

iv) Aqueous solution of $\mathrm{ZnSO_4}$ is added into the aqueous solution of $\mathrm{BaCl_2}$, this is the exaple of _____ reaction.

(a) displacement **(b)** double displacement

(c) redox **(d)** reduction

v) $n =$ _____. This law is also called as Snell's law.

(a) $\frac{\sin r}{\sin i}$ **(b)** $\frac{\sin r}{\sin e}$

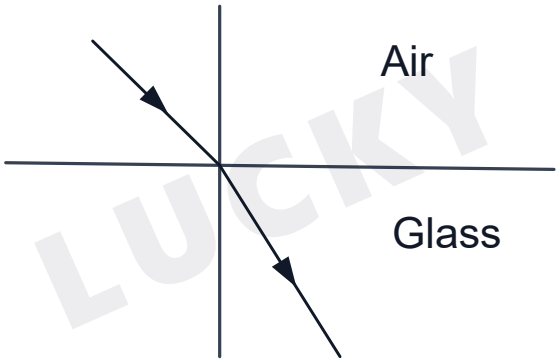
(c) $\frac{\sin e}{\sin r}$ **(d)** $\frac{\sin i}{\sin r}$

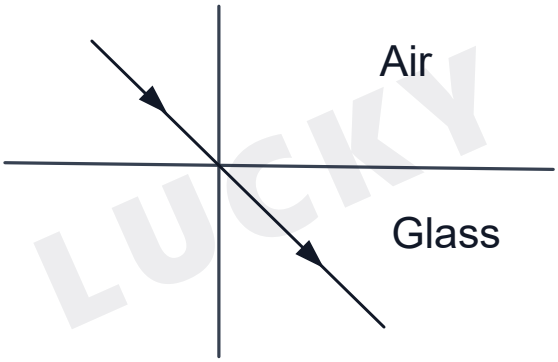
Q.1(B) **Answer the following sub-questions:** **[5 Marks]**

i) **Right or wrong sentence:**
As we go above the earth's surface, value of g increases.

ii) **Write the name.**
The element having stable electron configuration from the third period.

iii) **Identify the wrong figure from the following.**





iv) **State whether Right or Wrong:**
When the incident ray is parallel to the principal axis, the refracted ray passes through the principal focus.

v) **Right or wrong sentence.**
The speed of release of an object does not depend on the mass of the object.

Q.2(A) **Give scientific reasons (Any TWO):** **[4 Marks]**

i) It is reccommended to use air tight container for storing oil for long time.

ii) When the gas formed on heating limestone is passed through freshly prepared lime water, the lime water turns milky.

iii) It takes time for pieces of Shahabad tile to disappear in HCl, but its powder disappears rapidly.

Q.2(B) **Answer the following questions (Any THREE):** **[6 Marks]**

i) What is electrolysis?

ii) Identify which is the oxidising agent and reducing agent in the following reaction.

$$2\,H_2S + SO_2 \longrightarrow 3\,S\downarrow + 2\,H_2O$$

iii) **Explain the following reaction with the balanced equation.**
Reaction of zinc with dil hydrochloric acid

iv) **Identify from the following reaction the reactants that undergo oxidation and reduction.**

$$2\,Ag_2O \longrightarrow 4\,Ag + O_2\uparrow$$

v) What is corrosion?

Q.3 **Answer the following questions (Any FIVE):** **[15 Marks]**

i) Explain the factors affecting the rate of chemical reaction with example.

ii) What are reactants? Explain with the help of example.

iii) **Explain the following reaction with their balanced chemical equation.**
Ammonia gas reacts with hydrogen chloride.

iv) **Classify the following reaction as an exothermic or an endothermic reaction.**

a. Magnesium ribbon when burned in air, it forms magnesium oxide.

b. $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O + Heat}$

c. $\mathrm{2\,KClO_{3(s)} \xrightarrow{\Delta} 2\,KCl_{(s)} + 3\,O_2}$

d. $\mathrm{CaO + H_2O \longrightarrow Ca(OH)_2 + Heat}$

e. $\mathrm{CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_2}$

v) **Match the columns.**

	Reactants		Products	Types of chemical reaction
1.	$\mathrm{MgH_2}$	→	$\mathrm{Mg + H_2}$	Endothermic
2.	$\mathrm{2H_2S + SO_2}$	→	$\mathrm{3S + 2H_2O}$	Oxidation
3.	$\mathrm{CaO + H_2O}$	→	$\mathrm{Ca(OH)_2 + heat}$	Exothermic
				Redox

vi) Write three steps of writing chemical equations with example?

vii) What is the reaction called when oxidation and reduction take place simultaneously? Explain with one example.

viii) **Identify from the reaction the reactants that undergo oxidation and reduction.**

$$\mathrm{NiO + H_2 \longrightarrow Ni + H_2O}$$

Q.4	**Answer the following question (Any ONE):**	**[5 Marks]**

i) **Balance the given chemical reaction as per the instructions below.**

$$\mathrm{NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O}$$

a. Write names of reactants and products of chemical reaction.

b. Make a list of the elements in the chemical equation.

c. Write the number of atoms of reactants and products.

d. Balance the equation with proper coefficient and rewrite the equation.

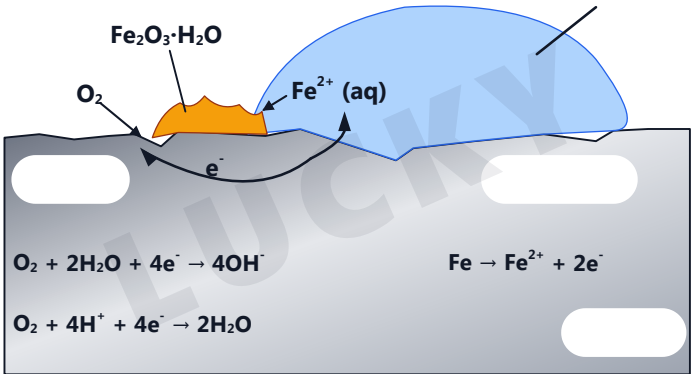
e. Oxidation means losing electrons then what is reduction?

ii) **Observe the following figure and write the answer of the question.**

a. Which process is shown in the figure?

b. Explain the chemical reaction shown in the figure.

c. Write the reactions on anode and cathode.



LUCKY

ALL THE BEST

DESIGNED BY YOUR MENTOR – LUCKY BHAIYA